Economics 345

Exploratory Data Analysis with R Fall 2026

Instructor: The instructor of record is Mark Jensen, who is an economist at the Federal Reserve Bank of Atlanta. The course is an initiative of the Federal Reserve Bank of Atlanta and is organized and taught by a group of analysts, research assistants, and economists from the Atlanta Fed. By the way we prefer to be called by our first name.

E-Mail: For official excuses, homework or project questions, or general course inquiries, e-mail *econ345.spelman@gmail.com*. This e-mail is maintained by a number of course leaders and will be checked periodically during business hours.

If you have something private, sensitive, time-critical, or confidential to discuss, please e-mail the instructor of record, Mark Jensen, at *Mark.Jensen@atl.frb.org*. This is Mark's work e-mail and he checks it regularly during business hours. Less so during non-business hours, so do not expect a quick response during the evening hourse or over the weekend.

Course Webpage: We will be using Canvas for all course material. The course Canvas webpage is: (TBD) All lectures and homework materials will be posted to Canvas. Students will submit assignments and receive grades in Canvas.

Lectures: Fridays from 9am-12pm in the 9th floor seminar room of the Federal Reserve Bank of Atlanta located on the corner of 10th and Peachtree in Midtown. To enter the bank you will need to pass through security. For those interning with the Survey Center you will be issued a security badge so you can enter without having to receive a guest pass from Law Enforcement. Until then Law Enforcement will issue you a temporary guest pass. This can take five to ten minutes so **please plan on arriving to the bank well before 9am** in order to pass through security and arrive to the seminar room in time to get your laptop connected to the wifi and to load the course Canvas page and RStudio.

Office Hours via Zoom: To be determined from the class questionnaire and availability of the teaching assistants. Plus by appointment (requests must be made at least a day in advance).

Course Syllabus

Required Text: *R for Data Science*, by Garrett Grolemund and Hadley Wickham (R4DS). R4DS is freely available online (https://r4ds.had.co.nz) and provides an excellent practical introduction to using R for data science. Hard copies are also available for purchase. R4DS provides insight into how to most effectively utilize R for data analysis. The author, Hadley Wickham, is also the author of many of the R packages used in the course.

Additional readings and videos will be posted on Canvas. You will get more out of the lecture and more out of the class by completing the assigned readings/videos, so they are included in the homework. The material is fair game for quizzes.

Required Software: Posit Cloud is a lightweight, cloud-based version of Rstudio – a web-based integrated development environment (IDE) for R – that allows anyone to do, share, and learn data science online. Posit Cloud can be run directly from your browser. There is nothing to configure and no dedicated hardware, installation, or purchase required. You will be receiving an e-mail with the Subject line "Mark Jensen has invited you to access a space on Posit Cloud". Simply click on link found in the e-mail and create an account. That's it.

Required Prerequisites: All students must have completed Econ 303 Econometrics with a grade of **B or higher**. This course does not require one to run linear regressions, instead, the course is focused on working with, vetting, plotting, and presenting data. No prior training in programming or data science is required.

Course Description

This course aims to give students the foundational data preparation and analysis skills needed to perform or support research in economics, finance, and related fields as well as analytics and policy analysis in business and government. Students will gain a fundamental understanding of data management and descriptive analysis, along with the statistical programming skills necessary to undertake these types of analyses using R, one of the leading statistical programming languages in data science, statistics, economics, and business. We will focus on developing the skills necessary to examine and effectively communicate relationships between variables, along with an exposure to the many potential pitfalls of data analysis (data analysis literacy).

Learning Objectives

- Reading, cleaning, and merging data sets
- Identifying and handling outliers along with other data issues
- Creating tables and charts to summarize data
- Analyzing and communicating relationships between variables of interest
- An introduction to how economic/statistical tools can be used for supplementary descriptive analysis

Grading

Student grades are based on the following components:

Final Project (worth 65% of the course grade)

- This course culminates in a final project and without the final project a student will not pass the course. Students will apply the coding skills learned in this course to explore economic questions using real world data and practice professional collaboration by working with a Atlanta Fed economist mentor and a project buddy. It presents an opportunity to experience work that is representative of advanced study in the field, and if done well, will produce an excellent writing sample and/or coding sample for job applications. Your relationship with your mentor and project buddy is a professional one so please treat it that way. In other words, failing to show up for a meeting with your mentor or analyst is not acceptable and will be reflected in your grade and/or further discpline. We understand that emergencies occur and when they do we ask that you to send a short e-mail to your mentor letting them know.

- The final project grade is based on 3 deliverables: a rough draft of the paper (graded on completeness), an in-class presentation, and a final paper. These 3 deliverables amount to 45%. The other 20% is comprised of project update assignments. To help make the project approachable and less overwhelming there are six graded project updates that amount to step-by-step instructions on how to complete your project. Completing the project updates on time will keep you on track to finishing your project. No such promise holds for those who do not keep up on the project updates.
- Separate detailed instructions and due dates will be posted on Canvas.

Weekly Homework (20%)

- Homework will be posted to Canvas. There are separate prompts for lecture-related homework and project-related homework, so we will often have two separate homeworks due at the same time – please check the schedule carefully. Students should submit all files to Canvas prior to the deadline, generally Thursday nights by midnight.
- Completing the homework on time is critical to your performance in this class, because the check-in quizzes and lectures build on the skills you used in the homework. The homework will also guide you through analyzing your real-world project data, which will help you make timely progress on your project.

  If you fall behind on your homework the lectures won't make any sense and neither will the next set of homework problems. This negative feedback loop will be impossible to escape from so **DO NOT FALL BEHIND ON YOUR HOMEWORK**. To ensure this does not happen plan right now on spending an hour outside of class for every hour in lecture, in other words, three hours a week studying and reviewing the material, and attending office hours. **Turning in late any homework during the first three weeks of class will seriously jeopardize your ability to pass the class.**
- We all have bad days - if you turn in all homeworks (reasonably completed!), your lowest grade will be dropped automatically.

Quizzes & Attendance (10%)

- Each class will start with a short in-class quiz to assess your understanding of the material covered in the previous homework, assigned readings/videos, and lecture. Quizzes begin promptly at 9am and end at 9:10am. You will receive a zero on quizzes that you do not take due to unexcused absence or lateness, and there will be no make-up.
- Again, we understand that we all have bad days and unexpected life events, so your lowest quiz will be dropped. Essentially, you may miss one class before your grade is affected. **Missing two of the first five classes will seriously jeopardize your ability to pass the class.**
- Students requesting an excused absence should e-mail *econ345.spelman@gmail.com* in advance and are responsible for taking their make-up quiz within 7 days of the missed class. Make-up quizzes will take place during the office hours sessions unless you have scheduled with us otherwise. **Even if your absence is unexcused, please let us know if you are going to miss class so that we aren't down at the security entrance waiting for you.**

- There will be an even shorter ungraded "exit ticket" at the end of class to assess our own performance as lecturers.

Communication, Timeliness, and Professionalism (5%)

- Students are expected to demonstrate professional communication and etiquette in all aspects of the course. The below criteria will be taken into consideration when determining the communication and professionalism component of the student's final grade.
- Coming prepared to class and to meetings with their project buddies and mentor. Being prepared for class includes having your camera on during meetings and sharing your screen with the teaching assistant during breakout sessions. Your engadgement in lectures and discussions is also a factor.
- Attending meetings; scheduling and canceling meetings with project buddies, TAs, and mentors, must be done with appropriate advance notice (usually 1 to 2 business days). Scheduling a meeting and then not showing up is unacceptable and if done more than once will result in a grade reduction.
- Respecting that course staff will check and respond to e-mails primarily during business hours (Monday-Friday, from 9am to 5pm).
- Using respectful and professional language when communicating with course staff.
- Anticipating your questions and needs, and consistenty communicating them professionally.

Late and Incomplete Policy

Late submissions of any assignments will be reduced by 2 percentage points for every partial hour late.[1]

Since the research project requires the generosity and time of the mentor and the project buddies, it is unfair to them for a students to receive an Incomplete in order to finish their project after the semester. Hence, a letter grade will be assigned at the end of the semester regardless of the circumstances.

At this point I wonder if you are actually reading this syllabus for its content. If you made it this far, for 5% extra credit on your first homework, email *econ345.spelman@gmail.com* (by 11:59pm August 30th) with the subject line "arrR! [your name]" and attach the poster from your favorite movie.

[1] For example, an assignment that earns an 85% that is submitted 2 hours and 20 minutes late will receive a 6 percentage point deduction, with a final score of 79%.

1 Tentative Study Guide

Tentative weekly schedule. See Canvas for the updated schedule and assigned readings/videos.

Date	Topic	Homework
Module 1: Fundamentals of Data Processing		
8/21	***No Class***	
8/28	Introduction, Posit Cloud Setup & R Basics	HW#1 (Due 9/3)
9/4	Data manipulation and the tidyverse	HW#2 (Due 9/10)
9/11	Summary statistics; Starting the project	HW#3 (Due 9/17) and Proj Update#1 (Due 9/19)
9/18	Code hygiene; R Markdown	HW#4 (Due 9/24) and Proj Update#2 (Due 9/26)
Module 2: Analyzing Relationships Between Variables		
9/25	Plots; dates/strings; tidyverse rev Research process; Project discussion	HW#5 (Due 10/1) and Proj Update#3 (Due 10/3)
10/2	Hypothesis testing & Correlation	HW#6 (Due 10/8) and Proj Update#4 (Due 10/10)
10/9	Loops; Debug; Proj discussion	HW#7 (Due 10/15) and Proj Update#5 (Due 10/17)
Module 3: Advanced Topics in Data Analysis		
10/16	***No Class*** – Homecoming	
10/23	Advance plots; Regression I;	HW#8 (Due 10/29) Proj Update#6 (Due 10/31)
10/30	Regression II & III; Proj Discussion	HW#9 (Due 11/5)
11/6	Coding rev; function programming	
11/13	Correlation vs Causation	**First draft of project due by 9am**
11/20	**Project presentations**	
11/27	***No Class*** – Thanksgiving Break	
12/11	**Final Version of Project Paper Due By 9am**	